INTERNSHIP PROJECT REPORT

ON

ENVIRONMENTAL MONITORING AND AIR QUALITY INTELLIGENCE

A project using Python-based data analysis, visualization, and machine learning

Submitted by

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DECLARATION

We hereby declare that the project report titled “Environmental Monitoring and Air Quality Intelligence” is prepared on the basis of the project notebook and the work represented in it. The report describes the data preparation, exploratory analysis, visualization, air-quality prediction, time-series feature engineering, and alert logic implemented in the notebook.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our internship organization, project mentor, and faculty members for providing us with the opportunity to work on Environmental Monitoring and Air Quality Intelligence. This project helped us understand how environmental datasets can be integrated and analyzed to identify pollution patterns, examine relationships between air quality and weather, and develop machine-learning-based estimation and forecasting workflows.

We also acknowledge the developers and maintainers of the Python libraries used in the notebook, including Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and Folium.

ABSTRACT

Environmental Monitoring and Air Quality Intelligence is a data-analysis project that combines daily air-quality records, weather observations, and city-level geographic metadata for Indian cities. The notebook uses Python to clean and integrate datasets, explore regional and city-level pollution patterns, visualize relationships between air-quality and meteorological variables, and develop Random Forest regression models. It also engineers lag and rolling-average features for time-series forecasting, creates a Folium-based geographic heat map, and implements rule-based air-quality alerts using AQI, wind speed, and precipitation conditions.

The first regression model reports an R^2 score of 0.8976 and mean absolute error of 14.79. The time-based forecasting model reports an R^2 score of 0.8178 and RMSE of 23.41. These results are notebook outputs and should be interpreted in the context of the dataset, selected features, and validation approach. The project demonstrates a practical workflow for environmental data integration and preliminary air-quality intelligence; it is not presented as an official public-health warning service.

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1. INTRODUCTION

Air quality is an important environmental concern because pollutant concentrations can vary across cities, regions, and seasons. Weather conditions such as wind speed, humidity, temperature, and precipitation may coincide with changes in pollutant levels and dispersion. A structured analysis of air-quality and weather data can help reveal patterns that are difficult to identify from individual records.

This project integrates air-quality observations with daily weather records and city metadata. It uses exploratory data analysis, charts, geographic visualization, and machine-learning models to study regional pollution patterns and estimate AQI values. It also prepares time-based features that can support forecasting experiments.

2. PROBLEM STATEMENT

Air-quality data, weather records, and geographic information are stored in separate datasets. Without combining these sources, it is difficult to compare pollution levels across regions, explore possible weather relationships, identify high-AQI locations, or prepare consistent features for prediction. The project addresses this data-analysis problem by creating an integrated dataset and applying visualization, regression, time-series feature engineering, and rule-based alert logic.

3. OBJECTIVES

- Identify and compare PM2.5 and PM10 pollution levels across regions and cities.
- Explore the relationship between AQI and selected weather variables, including wind speed, humidity, temperature, and precipitation.
- Prepare a consolidated dataset by joining air-quality, weather, and city metadata.
- Train a Random Forest regression model to estimate India AQI from pollutant, weather, and geographic features.
- Create lagged AQI values and rolling averages to support time-series forecasting.
- Visualize AQI intensity geographically and display city-level information.
- Demonstrate a rule-based alert function using AQI, wind speed, and precipitation thresholds.

4. SCOPE OF THE PROJECT

The project covers data loading, preprocessing, descriptive analysis, visual exploration, regression modeling, time-series feature engineering, interactive mapping, and a simple alert demonstration. The notebook focuses on daily records and city-level comparisons. The outputs can serve as a foundation for a future dashboard or a more rigorously validated environmental monitoring application.

The project does not establish causal relationships between weather and pollution, and its model predictions are not a substitute for official monitoring data or public-health guidance.

5. TOOLS AND TECHNOLOGIES

Tool / Library	Purpose in the project
Python	Main programming language for data analysis and model development.
Pandas	Loading CSV files, date conversion, joining datasets, grouping, and feature creation.
NumPy	Numerical operations and RMSE calculation.
Matplotlib	Plot creation and formatting.
Seaborn	Statistical charts, including bar plots and correlation heat maps.
Scikit-learn	Random Forest regression, train/test splitting, and model evaluation metrics.
Folium	Interactive map, heat-map layer, and city markers.
Google Colab / Jupyter-compatible notebook	Notebook-based execution and presentation of code and outputs.

6. DATASET DESCRIPTION

The notebook loads three CSV files:

Dataset	Role	Details visible in the notebook
cities.csv	City metadata	City, state, region, latitude, longitude, and elevation.
air_quality_daily.csv	Air-quality records	77,150 rows and 19 columns; includes AQI and pollutant fields.
weather_daily.csv	Weather records	124,500 rows and 14 columns; includes daily meteorological variables.

The air-quality and weather datasets are joined using date and city. The resulting records are then joined with city metadata using city. The integrated dataset contains 77,150 rows and 39 columns, according to the notebook output. Temporal fields for year, month, and day of week are derived from the date column.

7. METHODOLOGY AND IMPLEMENTATION

7.1 Loading the data

Pandas is used to read the three CSV files. Initial inspection displays sample city records and the dimensions of the air-quality and weather datasets. This provides a basic understanding of the data before integration.

7.2 Data cleaning and integration

The date fields in the air-quality and weather datasets are converted to datetime format. An inner join on date and city combines the daily air-quality and weather observations. A left join adds the state, region, coordinates, and elevation from the city metadata. The year, month, and day-of-week columns are extracted for time-based analysis.

7.3 Exploratory data analysis

The notebook calculates mean India AQI and mean PM2.5 for each region. It also groups records by city, calculates average India AQI, and prepares a bar chart of the ten cities with the highest mean AQI. This stage supports comparison of pollution levels across geographic areas.

7.4 Visualization

A correlation heat map is generated for India AQI, PM2.5, PM10, NO2, SO2, temperature, relative humidity, maximum wind speed, and precipitation. A monthly line chart shows average India AQI by calendar month. These visualizations help explore associations and seasonal patterns; correlation alone does not prove causation.

7.5 Random Forest AQI regression

The first model uses PM2.5, PM10, NO2, SO2, temperature, relative humidity, maximum wind speed, precipitation, elevation, latitude, and longitude as features. India AQI is the target. Rows with missing values in the selected model columns are dropped, and the data is split into training and testing sets using an 80:20 split with random_state=42. A RandomForestRegressor with 100 trees is trained. The notebook evaluates predictions using R^2 and mean absolute error and plots feature importance.

7.6 Time-series feature engineering and forecasting

Records are sorted by city and date. AQI lag features for the previous one, two, and three records are created for each city, together with seven-day rolling averages for AQI and PM2.5. After rows without the required lag values are removed, a second Random Forest model uses pollutant, weather, lag, and rolling-average features. Unlike the first model, this workflow uses an ordered 80:20 split to preserve temporal order in the train/test partition. RMSE and R^2 are reported.

7.7 Geographic visualization

Folium is used to create an interactive map centered on India. The notebook selects records from the latest date available in the merged dataset, adds an AQI-weighted heat-map layer, and places circle markers at city coordinates. Each marker displays the city, AQI, and AQI category.

7.8 Rule-based alert function

The function check_air_quality_alerts() retrieves the latest record for a selected city and prints AQI, AQI category, dominant pollutant, PM2.5, and wind speed. It adds a poor-air-quality message when AQI exceeds 200, a severe-health-hazard message when AQI exceeds 300, a stagnant-wind warning when wind speed is below 8.0 km/h, and a rain advisory when precipitation exceeds 10.0. These are the notebook's implemented rules and should be treated as demonstration thresholds, not official or clinically validated guidance.

7.9 Exporting the processed dataset

The notebook creates a latest-date national AQI summary and exports the integrated dataset to a CSV file named environmental_intelligence_processed.csv for possible downstream use in a dashboard or other application.

8. RESULTS AND OBSERVATIONS

8.1 Average pollution by region

Region	Average India AQI	Average PM2.5
Central	138.66	36.98
East	171.76	49.12
North	179.58	46.48
Northeast	125.13	26.93
South	115.90	22.87
West	129.76	31.04

In the notebook's regional summary, the North region has the highest average India AQI (about 179.58), followed by East (about 171.76). South has the lowest average AQI in this table (about 115.90). East has the highest average PM2.5 in this regional summary (about 49.12). These values describe the records in the integrated dataset and should not be generalized beyond its coverage without further validation.

8.2 Model evaluation

Model	Evaluation method	Reported result
Random Forest AQI regression	Random 80:20 train/test split; R^2 and MAE	$R^2 = 0.8976$; MAE = 14.79
Random Forest time-series forecasting	Ordered 80:20 split; R^2 and RMSE	$R^2 = 0.8178$; RMSE = 23.41

The first model reports an R^2 of 0.8976 and MAE of 14.79. The forecasting model reports an R^2 of 0.8178 and RMSE of 23.41. The metrics are not directly interchangeable because the models use different feature sets and split strategies. The notebook outputs do not include a full external validation study, confidence intervals, or a comparison against a simple baseline model.

8.3 Latest-date city summary

The notebook's latest-date summary is dated 25 October 2026. The ten highest AQI records shown begin with Mumbai (AQI 302, category Very Poor), Visakhapatnam (267, Poor), Kolkata (247, Poor), Prayagraj (222, Poor), and Lucknow (220, Poor). This is a snapshot of the notebook data, not a live measurement feed. The notebook also tests its alert function on Delhi for the same date, showing AQI 165 (Moderate), dominant pollutant O3, PM2.5 of 46.9 µg/m³, and wind speed of 8.5 km/h; under the coded rules, no alert is printed for that record.

8.4 Main project outputs

- Integrated dataset containing 77,150 rows and 39 columns.
- Regional average AQI and PM2.5 summary.
- Bar chart of the ten cities with the highest mean AQI.
- Correlation heat map for selected air-quality and weather variables.
- Monthly average AQI line chart.
- Feature-importance plot for the regression model.
- Interactive AQI heat map and city markers using Folium.
- Rule-based city alert report.
- Time-series lag and rolling-average features.
- Exported processed CSV file.

9. FUTURE SCOPE

- Use verified and documented data sources with clear coverage, units, and update frequency.
- Evaluate models with walk-forward or rolling-origin validation and compare them with simple baseline forecasts.
- Investigate feature leakage and ensure all features would be available at the intended prediction time.
- Add error analysis by city, region, season, and AQI range.
- Develop a dashboard to filter AQI trends by city, state, region, date, and pollutant.
- Connect the workflow to a reliable scheduled data pipeline, with data-quality checks and clear timestamps.
- Review alert thresholds against authoritative environmental standards and present appropriate source links and disclaimers.
- Consider uncertainty ranges and explainable model outputs to make predictions easier to interpret.

10. LEARNING OUTCOMES

- Loading and inspecting multiple CSV datasets using Pandas.
- Converting date fields and joining datasets on common keys.
- Creating time-based and rolling-window features.
- Using group-by summaries to compare regions and cities.
- Producing charts to explore trends and correlations.
- Training a Random Forest regression model and interpreting R^2, MAE, and RMSE.
- Creating a geographic visualization with Folium.
- Translating simple business or environmental rules into a Python alert function.
- Exporting processed data for future analysis or dashboard development.

11. CONCLUSION

The project demonstrates an end-to-end introductory workflow for environmental data analysis, from combining air-quality, weather, and geographic datasets to exploring pollution patterns and building regression and forecasting models. Regional summaries and visualizations make it easier to compare pollution levels, while the Folium map and rule-based alert function show how analytical results can be presented in a more accessible form.

The reported model scores indicate that the selected features contain useful predictive information in the notebook's evaluation setup, but further time-based validation, data quality checks, and threshold review would be necessary before using the workflow for operational decisions.

12. REFERENCES

- Project notebook supplied for this report: Environmental_Monitoring_Air_Quality_Intelligence.ipynb.
- Pandas documentation: https://pandas.pydata.org/docs/
- NumPy documentation: https://numpy.org/doc/
- Matplotlib documentation: https://matplotlib.org/stable/
- Seaborn documentation: https://seaborn.pydata.org/
- Scikit-learn documentation: https://scikit-learn.org/stable/
- Folium documentation: https://python-visualization.github.io/folium/